

#notes2

Operating System

An Operating System is core software that acts as an interface between user applications and computer hardware.

- * Manages system hardware and software resources
- * Provides common services for computer programs

Kernel

The kernel is the essential core component of an operating system.

- * Directly interacts with system hardware (CPU, RAM, storage)
- * Manages system resources and memory allocation

Additional OS Components

Aside from the kernel, an OS includes:

- * System utilities and command-line tools (shell)
- * System libraries
- * User interface (CLI or GUI)
- * Application software and package management tools

Linux

Linux is a free, open-source, Unix-like operating system kernel created by Linus Torvalds in 1991.

- * Combined with GNU tools to form a complete operating system
- * Highly scalable and used across various devices

Linux Distribution (Distro)

A Linux distribution is a complete OS built around the Linux kernel along with supporting software and tools.

- * Custom-packaged for specific user needs (e.g., Debian, Ubuntu, Fedora)
- * Includes default desktop environments and package managers

Linux Characteristics

- * ****Open Source:**** Source code is freely accessible, customizable, and shareable.
- * ****Multi-User & Multi-Tasking:**** Supports multiple users operating simultaneously and running multiple processes.
- * ****Security & Stability:**** Strong permission models reduce malware risks and maintain long system runtimes.
- * ****Portability:**** Able to run on diverse hardware platforms.

Debian

Debian is one of the oldest and most stable open-source Linux distributions.

- * Maintained by a global community of volunteers
- * Serves as the base foundation for distributions like Ubuntu

Licensing Agreements

- * ****Proprietary:**** Closed-source software owned by a single entity; source code is hidden (e.g., Windows).
- * ****Open Source:**** Source code is accessible to inspect, modify, and distribute.
- * ****Copyleft (GPL):**** Requires modified versions of the software to remain open-source under the same terms.
- * ****Permissive (MIT/BSD):**** Flexible open-source licensing with minimal restrictions on how code is reused.

Free Software & The 4 Freedoms

Free Software gives users full control over their computing through 4 essential freedoms:

- * ****Freedom 0:**** Run the program for any purpose.
- * ****Freedom 1:**** Study how the program works and change it.
- * ****Freedom 2:**** Redistribute copies to help others.
- * ****Freedom 3:**** Distribute modified versions to the community.

Virtualization

Virtualization is a technology that allows creating software-based representations of physical hardware.

- * Enables running multiple virtual machines on a single host computer
- * Allows running different operating systems simultaneously